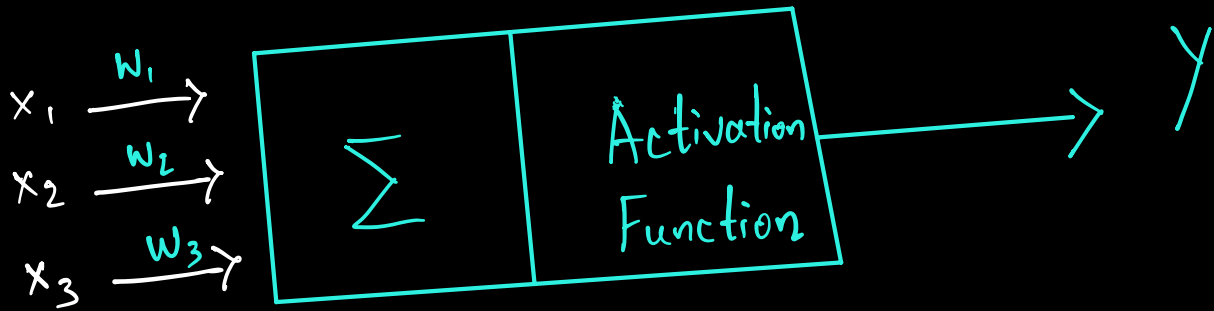


Perception Model



$$y = f(x)$$

$$y' = x_1 \cdot w_1 + x_2 \cdot w_2 + x_3 \cdot w_3$$

$$z = \underline{f(y')} \rightarrow \underline{(\text{Linear})}$$

Input layer

Dimension

Input Shape

Output layer

Regression
Classification

Binary

Multi

Hidden layer = Hyperparameters (2^n)

Loss Calculation

$$\text{Squared Error} = (\hat{y} - y)^2 = \text{Loss}$$

Diagram: A green arrow points from $\hat{y}$ to P (Predicted). A red arrow points from y to A (Actual).

Error :- Reducing the error $\rightarrow$ close to 0

Weights $\rightarrow$ Connections

$\hookrightarrow$ Learnable Parameters

$\hookrightarrow$ change w.r.t Time

W $\rightarrow$ Perfect values $\rightarrow$ Loss (close to 0)
Reaching Minima

$\{x, y\} \rightarrow$ Fixed

$\{w, b\} \rightarrow$ Tunable

Update the $\{w, b\}$

Gradient Descent $\rightarrow$ Optimization Technique

$$\{w \& b\} \approx 0(\text{loss})$$

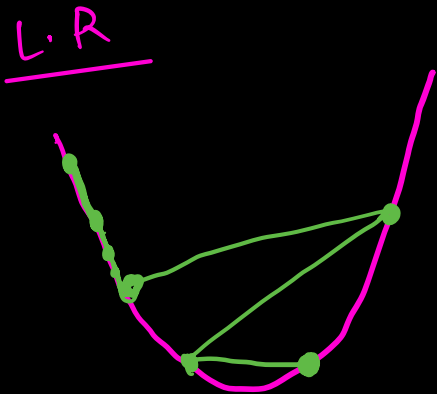
Weight Update Rule

$$W_{\text{new}} = W_{\text{old}} - \eta \left\{ \frac{\partial L}{\partial W_{\text{old}}} \right\}$$

η $\xrightarrow{\text{L.R (Learning Rate) } = 0.001}$

$\left\{ \frac{\partial E}{\partial W_{\text{old}}} \right\}$ $\rightarrow$ Calculated during Backprop

$\rightarrow$ Derivative of Error w.r.t W_{old}



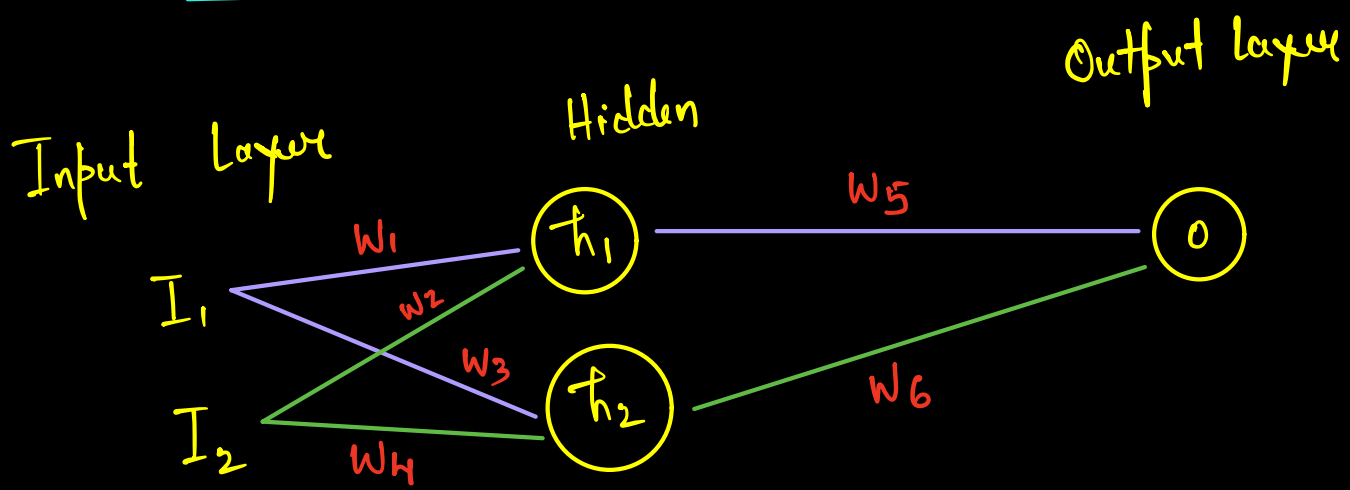
Bias Update

$$B_{\text{new}} = B_{\text{old}} - \eta \left\{ \frac{\partial E}{\partial B_{\text{old}}} \right\}$$

Learnable Params $\rightarrow \{w, b\}$

$$y = mx + \underline{b} \rightarrow \text{Intercept}$$

FORWARD PROPAGATION



$$w_1 = 0.11$$

$$w_2 = 0.21$$

$$w_3 = 0.12$$

$$w_4 = 0.08$$

$$w_5 = 0.14$$

$$w_6 = 0.15$$

$$x = \{2, 3\}$$

$$y = 1$$

Forward Pass $\rightarrow$ (Input $\rightarrow$ Hidden)

$$\begin{bmatrix} 2, 3 \end{bmatrix} \begin{bmatrix} 0.11 & 0.12 \\ 0.21 & 0.08 \end{bmatrix}$$

$$h_1 = 2 \times 0.11 + 3 \times 0.21 =$$

$$h_1 = 0.22 + 0.63 = 0.85$$

$$h_2 = 2 \times 0.12 + 3 \times 0.08 =$$

$$h_2 = 0.24 + 0.24 = 0.48$$

$$h_1, h_2 = [0.85, 0.48]$$

Hidden $\rightarrow$ Output

$$O = [0.85 \quad 0.48] \begin{bmatrix} 0.14 \\ 0.15 \end{bmatrix}$$

$$\begin{aligned} O &= 0.85 \times 0.14 + 0.48 \times 0.15 \\ &= 0.191 \end{aligned}$$

$$\text{Predicted} = 0.191$$

$$\text{Actual} = 1$$

Error Calculation

Loss Function = MSE

$$E = \frac{1}{2} (\hat{y} - y)^2$$

$$E = \frac{1}{2} (0.191 - 1)^2$$

$$E = 0.327$$

Neural Network

Epoch $\left\{ \begin{array}{l} 1 - \text{Forward Pass, calc (loss)} \\ 1 - \text{Backward Pass} \end{array} \right.$
→ Seen the entire dataset once.

1 Epoch = Network has seen the complete dataset once.
= 1 · FP + 1 · Backward
= FP + Backward Pass
5 Epoch = 5 (FP + BP)

Reverse Based

$$h_1 = i_1 w_1 + i_2 w_2$$

$$h_2 = i_1 w_3 + i_2 w_4$$

$$\text{Prediction} = (i_1 w_1 + i_2 w_2) \cdot w_5 + (i_1 w_3 + i_2 w_4) \cdot w_6$$

Back propagation

- 1) Reducing the Error
- 2) Update the weights.

→ optimization Technique
gradient Descent

Iterative Algorithm → Finding the minimum of a function.

Backpropagation

Weight update

$$\begin{aligned} * W_6 &= W_{6old} - \eta \left\{ \frac{\partial E}{\partial W_{6old}} \right\} \\ &\quad \cdot 0.001 \leftarrow \left\{ \frac{\partial E}{\partial W_{5old}} \right\} \\ W_5 &= W_{5old} - \eta \left\{ \frac{\partial E}{\partial W_{5old}} \right\} \end{aligned}$$

$$\frac{\partial E}{\partial w} = \frac{\partial E_{\text{move}}}{\partial \text{Prediction}} * \frac{\partial \text{Prediction}}{\partial \text{weight}}$$